

CAMILLE CHIU

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Updated: May 2026

EDUCATION

Harvard University, Cambridge, MA
Ph.D. in Astronomy

September 2027–

Yale University, New Haven, CT
B.S. in Astrophysics

September 2022–May 2026

College Station High School, College Station, TX

September 2017–May 2021

PUBLICATIONS

- **Chiu C.**, Chandra V., & Rix H.W., in prep. The Stellar Mass-Weighted Orbit and Metallicity Distribution of the Milky Way. *In preparation for submission.*
- **Chiu C.**, Li Y., Huber D., Ong J., van Saders J., & Crawford C., in prep. Constraining Mass Loss Across the Red Giant Branch Using Wide Binaries from Gaia DR3. *In preparation for submission.*
- **Chiu C.**, Geha M., Cerny W., Kallivayalil N., Richstein H., Garling C., & Willman B. (2026). Willman 1 Revisited: The Kinematics, Chemistry, and Orbital Properties of a Potentially-Disrupting Dwarf Galaxy. *The Astrophysical Journal*, 1002, 55.

RESEARCH EXPERIENCE

The Stellar Mass-Weighted Orbit and Metallicity Distribution of the Milky Way
Summer Research Intern under the mentorship of Vedant Chandra & Prof. Hans-Walter Rix

Summer 2025-present
Heidelberg, Germany

- Developed a new method for reconstructing the orbit and chemical stellar mass distribution of the Milky Way by modeling the *Gaia* selection function
- Used this model to calculate fundamental structural properties of the Milky Way to situate our Galaxy in the context of galactic formation/chemical evolution models as well as the local galaxy population

Constraining Mass Loss on the Red Giant Branch Using Wide Binaries and Asteroseismology

NSF REU Researcher under the mentorship of Dr. Yaguang Li & Prof. Dan Huber

Summer 2024-present
Honolulu, HI

- Measured RGB mass loss for a sample of nine red clump+subgiant wide binaries by fitting each binary to stellar evolution models using observational, asteroseismic, and coevolution constraints

Willman 1 Revisited: The Kinematics, Chemistry, and Orbital Properties of a Potentially-Disrupting Dwarf Galaxy

Yale Undergraduate Researcher under the mentorship of Prof. Marla Geha

Spring 2023-Fall 2025
New Haven, CT

- Developed an original framework to differentiate between Milky Way foreground interloper stars and members of Willman 1 (W1)
- Used member sample to characterize W1's kinematic, metallicity, and mass segregation profiles as well as perform dynamical modeling to determine the orbital history of W1
- Concluded that W1 is a dwarf galaxy that is likely undergoing tidal disruption; we caution that the inferred dynamical mass from its internal kinematics does not accurately reflect the actual mass of the system

AWARDS/FELLOWSHIPS

George Beckwith Prize, to the undergraduate most proficient in some branch of astronomy

May 2026

Yale Science & Engineering Association (YSEA) Senior Distinction Award

April 2026

National Goldwater Scholarship (\$7,500)

March 2025

Best Poster Presentation Award (University of Connecticut Conference for Women in Physics)

January 2025

Chambliss Astronomy Achievement Student Award (245th AAS Meeting)

January 2025

University of Hawai'i Institute for Astronomy REU Fellow (\$7,000)	Summer 2024
Yale First-Year Summer Research Fellowship in Science & Engineering (\$4,300)	Summer 2023
International Science and Engineering Fair (ISEF)	
• Finalist, selected as one of six best in fair projects to represent the Austin district	May 2019, 2020, 2021
• Honorable Mention from the National Aeronautics Space Administration (NASA)	May 2019
• 4th place in Astronomy/Physics (\$500)	May 2021

CONFERENCES/TALKS

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| • Attendee at the 247th Meeting for the American Astronomical Society in Phoenix, AZ
<i>Poster: The Stellar Mass-Weighted Orbit and Metallicity Distribution of the Milky Way</i> | January 2026 |
| • Yale University Undergraduate Kick-off Event
<i>Talk: The Stellar Mass-Weighted Orbit and Metallicity Distribution of the Milky Way</i> | September 2025 |
| • MPA Milky Way Group Meeting
<i>Talk: The Stellar Mass-Weighted Orbit and Metallicity Distribution of the Milky Way</i> | August 2025 |
| • Attendee at APS Conference for Undergraduate Women in Physics at the University of Connecticut
<i>Poster: The Kinematics of Willman 1: A Disrupting Dwarf Galaxy</i> | January 2025 |
| • Attendee at the 245th Meeting for the American Astronomical Society in National Harbor, MD
<i>Poster: Constraining Mass Loss on the Red Giant Branch Using Wide Binaries and Asteroseismology</i> | January 2025 |
| • Yale University Undergraduate Kick-off Event
<i>Talk: Constraining Mass Loss on the Red Giant Branch Using Wide Binaries and Asteroseismology</i> | September 2024 |
| • University of Hawai'i REU Final Presentations
<i>Talk: Constraining Mass Loss on the Red Giant Branch Using Wide Binaries and Asteroseismology</i> | August 2024 |
| • Attendee at APS Conference for Undergraduate Women in Physics at CUNY
<i>Poster: Re-assessing Milky Way Dwarf Galaxy Willman 1</i> | January 2024 |
| • Invited attendee at Dwarfs in the Local Group and Beyond Regional Conference at the Flatiron CCA | July 2023 |

OUTREACH

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| • Research group leader for Warrior Scholars Program STEM bootcamp | Summer 2026 |
| • President the Yale chapter of Women and Gender Minorities in Physics (WiP+) | Fall 2025-Spring 2026 |
| • Yale Astronomy Department AstroSibs Undergraduate Coordinator | Spring 2025-Spring 2026 |
| • Writer for the Yale Scientific Magazine | Spring 2025-Spring 2026 |
| • Regular volunteer for Yale Pathways events (e.g. Girls' Science Investigations) | Fall 2022-Spring 2026 |
| • Independent project: Khroma website
– created a website that guides the user through the creation of astronomical RGB color images from JWST data | Spring 2024 |
| • High school tutoring (physics) | Spring 2024 |
| • Yale Prison Education Initiative (YPEI) volunteer | Summer 2023 |
| • Invited speaker at the Austin Astronomical Society, <i>Talk: Galactic Habitability</i> | August 2020 |

SKILLS

- **Data Science/Programming:** advanced Python; intermediate LaTeX, R; basic Mathematica, C++
- **Foreign languages:** French (C1 level, spent gap year between high school and college as an exchange student in France, attending French lycée and living with a host family)
- **Memberships:** Society for Physics Students (2022-2026), American Astronomical Society (2024-present), Sigma Pi Sigma Honor Society (2026-present)

OTHER INTERESTS

- **Ballet,** current board member of the Yale Ballet Company
- **Music,** play piano as background music, as an accompanist, and for personal enjoyment; currently teach local elementary schooler piano basics; member of the Yale Handbells Ensemble (former co-president)
- **Reading,** science fiction, philosophy, social sciences (Ted Chiang, Elena Ferrante, Garth Greenwell, Albert Camus)